

A small study of page layout

Two hundred readers read the same twelve documents twice, once in the original layout and once after it was reset with a single column, wider margins and fewer words per line. Every measure below compares the two readings.

The two readings were a month apart, so that readers had forgotten most of each document, and half the readers met the reset layout first.

Results

Table 1. Results of the study

Measure	Before	After	Change
Reading time per page	84 s	61 s	-27%
Readers who lost their place	41	16	-61%
Questions answered correctly	72%	88%	+16 pts
Pages reread at least once	2.9	1.4	-52%
Words per line	13	9	-31%
Satisfaction score	6.1	8.3	+2.2

The largest change was in readers who lost their place, which fell by more than half. Reading time per page fell by about a quarter. Satisfaction rose by two points on a ten point scale, the smallest change in relative terms and the one readers mentioned first when asked.

Every change in the table points the same way. No measure got worse after the reset, and none of the differences was close to the margin of error.

Words per line is the one measure the reset set directly. The others followed from it, which is why the study treats line length as the cause and the rest as effects.

What the numbers do not say

The table reports averages. Behind the reading time of 61 seconds per page after the reset sits a wide spread: the fastest readers finished in half that, and the slowest took twice as long. The 88 percent of questions answered correctly hides that two of the twelve documents were answered no better than before.

A table is a good home for a number and a poor home for a caveat. When a passage is retrieved from this document, a row of the table carries its label and its values together only if the parser kept the row together. A parser that reads the cells as loose words leaves the number 61 with nothing to say what it measures.

Averages also hide who gained. Readers who read slowly to begin with gained the most from the reset; the fastest readers gained almost nothing, because they rarely lost their place in either layout.

The satisfaction score came from one question at the end of each document: how easy was this to read, from one to ten. It measures what readers felt, not what they did, and the two do not always agree.

Some changes interact. Fewer words per line means more lines per page and so more pages, and a reader who turns more pages has more chances to lose their place at the top of a new one. The reset still won, but by less than it would have without the extra pages.

Finally, a month between readings is long enough to forget details but not the gist. Some of the gain in correct answers may come from meeting the document a second time, which is why half the readers saw the reset layout first.

None of this undoes the result. It says how far the numbers can be trusted, and which questions the study cannot answer on its own.

Results by document

Table 2 gives the reading time per page for each of the twelve documents in both layouts. The difference is the extra time two columns cost, in seconds per page.

Table 2. Reading time per page for each document

Document	One column	Two columns	Difference
Bus timetable guide	40 s	55 s	+15 s
Tenancy agreement	92 s	104 s	+12 s
Recipe booklet	35 s	37 s	+2 s
Council newsletter	48 s	47 s	-1 s
Pension statement	71 s	86 s	+15 s
Product manual	66 s	79 s	+13 s
Insurance policy	88 s	101 s	+13 s
School report	44 s	52 s	+8 s
Energy bill	39 s	46 s	+7 s
Train ticket terms	57 s	69 s	+12 s
Library leaflet	33 s	38 s	+5 s
Job advert	41 s	50 s	+9 s

The council newsletter is the only row where two columns were not slower. Its stories were short and stood alone, so readers rarely had to jump between columns.

The rows look alike on purpose: same columns, same units, numbers in the same range. That is what most tables in real reports look like.

Longer documents cost more in two columns. The tenancy agreement and the insurance policy, the two longest, also had among the biggest differences once their length is taken into account.

The recipe booklet and the library leaflet were the shortest, and two columns cost them almost nothing.

A reader who wants one number from this table needs both the row label and the column heading. Lose either, and the number means nothing.

The changes at a glance

The chart below shows the same six measures as Table 1, each as a change from its value before the reset. Bars below the line are measures that fell; bars above it are measures that rose.

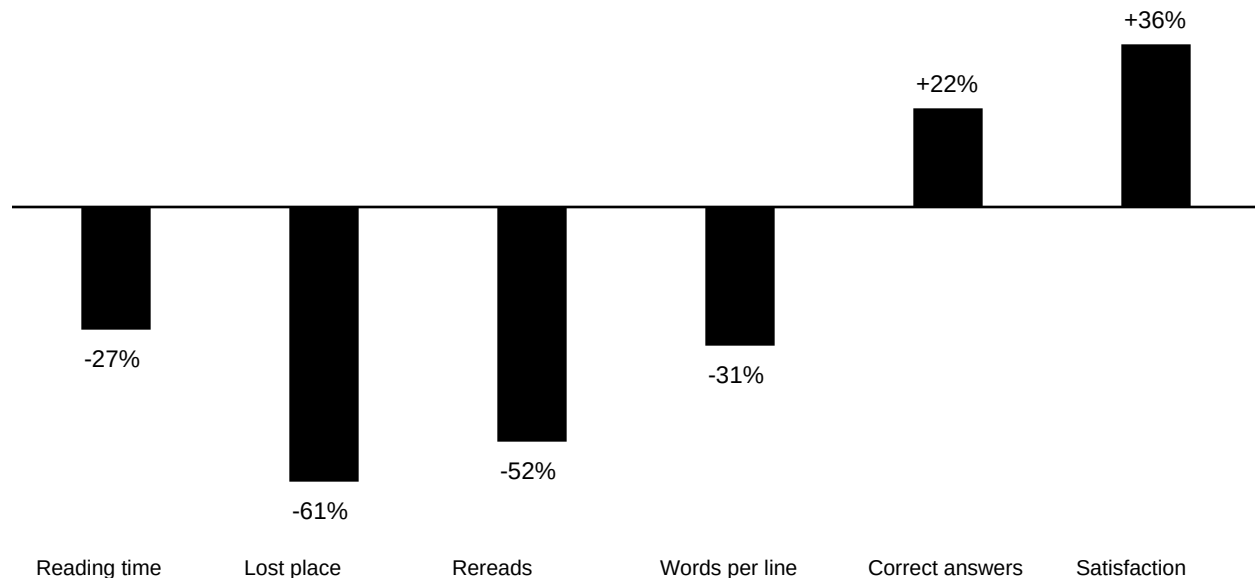


Figure 1. Change in each measure after the reset, as a percentage of the value before.

The chart makes the pattern easier to see than the table does: everything that costs a reader effort went down, and everything a reader gains went up.

A chart like this is a picture. A parser reads its caption and its labels as text, but the heights of the bars are only lines on the page, so the numbers behind them are lost unless they are also written down, as they are in Table 1.

That is why a question about the chart can only be answered from its caption, or from the table it was drawn from.

The two rising bars are the measures a reader would want to rise: answers that were right, and how easy the documents felt to read.

The four falling bars are all kinds of effort: time on the page, losing one's place, rereading, and the length of each line.

Summary for the board

This page was printed, signed and scanned before the report was sent out, so it is a picture of a page with no text inside it.

Across all twelve documents the single column saved an average of nine seconds a page, which over a report of forty pages is six minutes for every reader.

The board approved the reset layout for every report the council publishes from next spring, and asked for the study to be repeated after a year.

The study took four months from the first reader to this summary, and cost less than printing one annual report.

Questions from the board about the method are answered in the notes at the end of the report.

A copy of this summary, with the signatures, is kept with the minutes of the meeting that approved it.

Notes and caveats

The readers were volunteers recruited through the council's newsletter, so they were people who already read council documents and may have been keener than most.

Each reading was timed by the screen software, not by a person with a stopwatch, so the times are exact to the second but include any pause a reader took.

The questions after each document were the same in both readings. A reader who remembered a question from the first reading may have looked for its answer in the second, which would favour whichever layout came second.

The twelve documents were chosen to be ordinary rather than difficult. Technical manuals and legal texts with dense cross-references were left for a later study.

Three readers did not finish the second reading and were left out entirely, rather than counted for the documents they did finish.

The printed copies used for the signed summary were set in the reset layout. The study itself was read entirely on screen.

All the figures in this report are rounded. A change shown as a quarter or a half is the nearest simple fraction to the measured value.

The reading times in Table 2 are averages over all two hundred readers. The spread within each document was about as wide as the spread described on page 2.

Satisfaction was asked once per document and reading, not once per page, so it cannot be compared with the times per page directly.

The study did not track eye movements. Where it says a reader lost their place, it means the screen software saw them scroll back up by more than three lines.

No reader saw the same document in the same layout twice.

The council paid for the study, but the analysis was done by an outside team who had no say in which layout was chosen.